Exercises from Ch. 5: Algebraic Extensions, Lang

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Exercises

Exercise 0.1. (Exercise 2)
Let $E = F(\alpha)$ where α is algebraic over F, of odd degree. Show that $E = F(\alpha^2)$.

Proof. Assume that $E = F(\alpha)$ where α is algebraic of odd degree. Hence, $[F(\alpha)\colon F] = 2k+1$ with $k \in \mathbb{Z}_{\geq 0}$. Observe that we have the following tower

$$F \subseteq F(\alpha^2) \subseteq F(\alpha)$$

since $\alpha^2 \in F(\alpha)$ ($F(\alpha)$ is closed). Let $f(X) = X^2 - \alpha^2 \in F(\alpha^2)[X]$. Hence, $f(\alpha) = 0$. Thus, $[F(\alpha)\colon F(\alpha^2)] \leq 2$. By the tower law,

$$[F(\alpha)\colon F] = \underbrace{[F(\alpha)\colon F(\alpha^2)]}_{\leq 2} \cdot [F(\alpha^2)\colon F]$$

We see that for the right hand side of this equation to be odd, we cannot have $[F(\alpha)\colon F(\alpha^2)] = 2$. Thus, $[F(\alpha)\colon F(\alpha^2)] = 1 \iff F(\alpha) = F(\alpha^2)$.

□

Exercise 0.2. (Exercise 3)
Let α and β be two elements which are algebraic over F. Let $f(X) = \mathrm{Irr}(\alpha, F, X)$ and $g(X) = \mathrm{Irr}(\beta, F, X)$. Suppose that $\deg f$ and $\deg g$ are relatively prime. Show that g is irreducible in the polynomial ring $F(\alpha)[X]$.

Proof. Set $x = \deg f$ and $y = \deg g$. BWOC assume that g is reducible in $F(\alpha)[X]$. Hence,

$$[F(\alpha,\beta)\colon F(\alpha)] = z < \deg g \quad (z \geq 1).$$

By the tower laws,

$$[F(\alpha,\beta)\colon F] = \underbrace{[F(\alpha,\beta)\colon F(\alpha)]}_{=z} \cdot \underbrace{[F(\alpha)\colon F]}_{=x}$$

and

$$[F(\alpha,\beta)\colon F] = [F(\alpha,\beta)\colon F(\beta)] \cdot \underbrace{[F(\beta)\colon F]}_{=y}$$

Set $l = [F(\alpha,\beta)\colon F(\beta)]$. Thus, $zx = yl$ $(y \big| zx)$. Since $(x,y) = 1$, there exists $u, v \in \mathbb{Z}$ such that

$$ux + vy = 1$$

Multiplying both sides by z,

$$
\begin{aligned}
(zx)u + (zy)v &= z \\
ylu + zyv &= z \\
y(ly + zv) &= z
\end{aligned}
$$

Hence, $y\big|z$, but $z < y$. Therefore, $g(X)$ is irreducible in $F(\alpha)[X]$.

$\square$

Exercise 0.3. (Exercise 4)
Let α be the real positive fourth root of 2. Find all intermediate fields in the extension $\mathbb{Q}(\alpha)$ of $\mathbb{Q}$.

Proof. Let K be an intermediate field, i.e.

$$\mathbb{Q} \subseteq K \subseteq \mathbb{Q}(\alpha).$$

By the tower law, we have that

$$[K\colon \mathbb{Q}]\Big|[\mathbb{Q}(\alpha)\colon \mathbb{Q}] \Rightarrow [K\colon \mathbb{Q}]\Big|4.$$

Then all non-trivial intermediate fields of this extension are quadratic, that is, $[K\colon \mathbb{Q}] = 2$, and so $[\mathbb{Q}(\alpha)\colon K] = 2$. Since $\{1, \alpha, \alpha^2, \alpha^3\}$ is a basis for $\mathbb{Q}(\alpha)/\mathbb{Q}$, we have $K = \mathbb{Q}(\alpha^2)$ with basis $\{1, \alpha^2\}$. Therefore, $\mathbb{Q}(\alpha^2)$ is the only intermediate field in the extension $\mathbb{Q}(\alpha)$ of $\mathbb{Q}$. Explicitly,

$$\mathbb{Q} \subseteq \mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\sqrt[4]{2}).$$

$\square$

Exercise 0.4. (Exercise 6)
Show that $\sqrt{2} + \sqrt{3}$ is algebraic over $\mathbb{Q}$, of degree 4.

Proof. We will show that $\sqrt{2} + \sqrt{3}$ is algebraic by finding some $f(X) \in \mathbb{Q}[X]$ such that $f(\sqrt{2} + \sqrt{3}) = 0$. Let $\alpha = \sqrt{2} + \sqrt{3}$. Then,

$$
\begin{aligned}
\alpha^2 &= 5 + 2\sqrt{6} \\
\alpha^2 - 5 &= 2\sqrt{6} \\
\alpha^4 - 10\alpha^2 + 25 &= 24 \\
\alpha^4 - 10\alpha^2 + 1 &= 0
\end{aligned}
$$

Thus, take $f(X) = X^4 - 10X^2 + 1 \in \mathbb{Q}[X]$. By construction, $f(\alpha) = 0$ and thus α is algebraic over $\mathbb{Q}$. Moreover $\mathbb{Q}(\sqrt{2} + \sqrt{3}) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3})$. But also

$$(\sqrt{2} + \sqrt{3})(\sqrt{2} - \sqrt{3}) = -1,$$

so $\sqrt{2} - \sqrt{3} \in \mathbb{Q}(\sqrt{2} + \sqrt{3})$, and hence

$$\sqrt{2} = \frac{1}{2}((\sqrt{2} + \sqrt{3}) + (\sqrt{2} - \sqrt{3})) \in \mathbb{Q}(\sqrt{2} + \sqrt{3})$$

and likewise $\sqrt{3} \in \mathbb{Q}(\sqrt{2} + \sqrt{3})$. Therefore, $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Thus, we have

$$[\mathbb{Q}(\sqrt{2} + \sqrt{3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4.$$

We found a polynomial $f(X)$ of degree 4 and thus conclude that we also have $f(X) = \mathrm{Irr}(\sqrt{2} + \sqrt{3}, \mathbb{Q}, X)$. $\square$

Exercise 0.5. (Exercise 8)
Let $f(X) \in k[X]$ be a polynomial of degree n. Let K be its splitting field. Show that $[K\colon k]$ divides $n!$.

Proof. We proceed by induction. If $n = 1$, then k itself is the splitting field of $f(X)$ and $(1 =)[k\colon k]\big|1!$. Suppose that we can find a splitting field of dimension dividing $\deg f\,!$ over k for all $f(X) \in k[X]$ such that $\deg f < n$.
First, suppose that $f(X)$ is irreducible over k. Consider the extension $k(a_1) \supseteq k$ where a_1 is a root of $f(X)$. Hence,

$$f(X) = (X - a_1)g(X) \in k(a_1)[X]$$

with $\deg g = n - 1$. Thus, by the inductive hypothesis, there is a splitting field E for $g(X)$ over $k(a_1)$ such that $[E\colon k(a_1)]\big|(n-1)!$. Then, $E = k(a_1)(a_2, \ldots, a_n)$ where $a_2, \ldots, a_n$ are the roots of $g(X)$. Since $f(X)$ splits over $E = k(a_1, \ldots, a_n)$, E is a splitting field for $f(X)$ over k. Since $[k(a_1)\colon k] = n$ (since $f(X)$ is irreducible over k), $[E\colon k] = n \cdot [E\colon k(a_1)]$, and this divides $n!$ because $[E\colon k(a_1)]\big|(n-1)!$.
Now, secondly, suppose $f(X)$ is reducible in $k[X]$, i.e. $f(X) = g(X)h(X)$, where $\deg g = m < n$ and $\deg h = l < n$, then by the inductive hypothesis there is a splitting field $E = k(a_1, \ldots, a_m)$ for $g(X)$ over k such that $[E\colon k]\big|m!$. Again, by the inductive hypothesis, there is a splitting field $L = k(b_1, \ldots, b_l)$ for $h(X)$ over E such that $[E\colon L]\big|l!$. Then, $k(a_1, \ldots, a_m, b_1, \ldots, b_l) = K$ is a splitting field for $f(X)$ over k and $[K\colon k] = [K\colon E] \cdot [E\colon k]$ which divides $m!k!$, which divides $n!$. (Observe that $n!/m!k! = n!/m!(n-m)!$ is an integer, nCm).

□

Exercise 0.6. (Exercise 9)
Find the splitting field of $X^{p^8} - 1$ over the field $\mathbb{Z}/p\mathbb{Z}$.

Proof. Let $f(X) = X^{p^8} - 1 \in \mathbb{Z}/p\mathbb{Z}[X]$. Since we are in characteristic p, $f(X) = X^{p^8} - 1 = (X-1)^{p^8}$. Thus, $f(X)$ splits over $\mathbb{Z}/p\mathbb{Z}$. Note that 1 is a root of multiplicity p^8, thus $f(X)$ is not separable.

□

Exercise 0.7. (Exercise 10)
Let α be a real number such that $\alpha^4 = 5$.

a) Show that $\mathbb{Q}(i\alpha^2)$ is normal over $\mathbb{Q}$.

b) Show that $\mathbb{Q}(\alpha + i\alpha)$ is normal over $\mathbb{Q}(i\alpha^2)$.

c) Show that $\mathbb{Q}(\alpha + i\alpha)$ is not normal over $\mathbb{Q}$.

Proof. a) We have $\alpha^2 = \sqrt{5}$. Thus, we want to show $\mathbb{Q}(i\sqrt{5})/Q$ is normal. The irreducible polynomial of $i\alpha^2$ over $\mathbb{Q}$ is $\mathrm{Irr}(i\alpha^2, \mathbb{Q}, X) = X^2 + 5 = f(X)$. Then $f(\pm i\sqrt{5}) = 0$. Thus, $f(X) = (X + i\sqrt{5})(X - i\sqrt{5})$. Hence, $\mathbb{Q}(i\alpha^2)$ is the splitting field for $f(X)$. Therefore, $\mathbb{Q}(i\alpha^2)/\mathbb{Q}$ is normal.

b) Let $f(X) = X^2 - 2i\sqrt{5} \in \mathbb{Q}(i\alpha^2)[X]$. Then $f(X) = (X + (\sqrt[4]{5} + i\sqrt[4]{5}))(X - (\sqrt[4]{5} + i\sqrt[4]{5}))$ in $\mathbb{Q}(\alpha + i\alpha)$. Hence, $\mathbb{Q}(\alpha + i\alpha)$ is the splitting field for $f(X)$. Thus, $\mathbb{Q}(\alpha + i\alpha)/\mathbb{Q}(i\alpha^2)$ is normal.

c) Consider $X^4 + 20 \in \mathbb{Q}[X]$ (one can derive this from $\beta = \sqrt[4]{5} + i\sqrt[4]{5}$ and applying arithmetic to until you get something completely in $\mathbb{Q}$). Thus, $f(X) = (X^2 + 2i\sqrt{5})(X^2 - 2i\sqrt{5}) = (X + (\sqrt[4]{5} + i\sqrt[4]{5}))(X - (\sqrt[4]{5} + i\sqrt[4]{5}))(X^2 - 2i\sqrt{5})$. Observe that $X^2 - 2i\sqrt{5} = (X + (\sqrt[4]{5} - i\sqrt[4]{5}))(X - (\sqrt[4]{5} - i\sqrt[4]{5}))$, but $\pm(\sqrt[4]{5} - i\sqrt[4]{5})$ is not necessarily in $\mathbb{Q}(\sqrt[4]{5} + i\sqrt[4]{5}) = \mathbb{Q}(\alpha + i\alpha)$. We argue this. By way of contradiction, suppose $\alpha - i\alpha \in \mathbb{Q}(\alpha + i\alpha)$. Then $\alpha = \frac{1}{2}[(\alpha + i\alpha) + (\alpha - i\alpha)]$ and $i\alpha = \frac{1}{2}[(\alpha + i\alpha) - (\alpha - i\alpha)]$. This would imply that $i \in \mathbb{Q}(\alpha + i\alpha)$ which is impossible since $\mathbb{Q}(\alpha + i\alpha)$ is generated by a single complex number with $\alpha \in \mathbb{R} \setminus \{0\}$. Hence, $\alpha - i\alpha \notin \mathbb{Q}(\alpha + i\alpha)$. Thus, since this polynomial doesn't completely split over this extension despite having a root, we conclude that $\mathbb{Q}(\alpha + i\alpha)/\mathbb{Q}$ is not normal.

□

Exercise 0.8. (Exercise 11)
Describe the splitting fields of the following polynomials over $\mathbb{Q}$, and find the degree of each such splitting field.

(a) $X^2 - 2$ (b) $X^2 - 1$
(c) $X^3 - 2$ (d) $(X^3 - 2)(X^2 - 2)$
(e) $X^2 + X + 1$ (f) $X^6 + X^3 + 1$
(g) $X^5 - 7$

Proof. a) The splitting field for X^2-2 over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{2})$. Thus, $[\mathbb{Q}(\sqrt{2})\colon \mathbb{Q})]=2$.

b) Observe that $X^2-1=(X+1)(X-1)\in\mathbb{Q}[X]$. Therefore $\mathbb{Q}$ is the splitting field and thus $[\mathbb{Q}\colon\mathbb{Q}]=1$.

c) The roots of $X^3-2\in\mathbb{Q}[X]$ consists of one real root and two complex roots, namely $\sqrt[3]{2},\zeta\sqrt[3]{2}$, and $\zeta^2\sqrt[3]{2}$ with $\zeta\in\mu_3$ a third root of unity (in this case primitive). Also, $\zeta^2=-\zeta-1$ and thus $\mathbb{Q}(\sqrt[3]{2},\zeta)\supseteq\mathbb{Q}$ is the splitting field of X^3-2. Then from $\mathbb{Q}(\sqrt[3]{2})\cap\mathbb{Q}(\zeta)=\mathbb{Q}$ and $\mathbb{Q}\subseteq\mathbb{Q}(\sqrt[3]{2})\subseteq\mathbb{Q}(\sqrt[3]{2},\zeta)$, we have

$$[Q(\sqrt[3]{2},\zeta)\colon\mathbb{Q}]=[\mathbb{Q}(\sqrt[3]{2},\zeta)\colon\mathbb{Q}(\sqrt[3]{2})]\cdot[\mathbb{Q}(\sqrt[3]{2})\colon\mathbb{Q}]=2\cdot3=6.$$

d) Let $f(X)=g(X)h(X)=(X^3-2)(X^2-2)\in\mathbb{Q}[X]$. Then by part (a) and (c), the splitting fields of $g(X)=X^3-2$ and $h(X)=X^{-}2$ over $\mathbb{Q}$ are $\mathbb{Q}(\sqrt[3]{2},\zeta)$ and $\mathbb{Q}(\sqrt{2})$, respectively. Then since $\mathbb{Q}(\sqrt[3]{2},\zeta)\cap\mathbb{Q}(\sqrt{2})=\mathbb{Q}$, the splitting field of $f(X)=(X^3-2)(X^2-2)$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[3]{2},\zeta)\mathbb{Q}(\sqrt{2})=\mathbb{Q}(\sqrt[3]{2},\zeta,\sqrt{2})$. Then by the tower law,

$$[\mathbb{Q}(\sqrt[3]{2},\zeta,\sqrt{2})\colon\mathbb{Q}]=[\mathbb{Q}(\sqrt[3]{2},\zeta,\sqrt{2})\colon\mathbb{Q}(\sqrt{2})]\cdot[\mathbb{Q}(\sqrt{2})\colon\mathbb{Q}]=6\cdot2=12.$$

e) Note that $\mathrm{Irr}(\zeta,\mathbb{Q},X)=X^2+X+1=\Phi_3(X)$ and thus the splitting field is $\mathbb{Q}(\zeta)$ with $[\mathbb{Q}(\zeta)\colon\mathbb{Q}]=\deg\Phi_3(X)=\varphi(3)=2$.

f) Observe that $f(X)=X^6+X^3+1=\Phi_3(X^3)=\Phi_{3^2}(X)=\Phi_9(X)$. Thus $f(X)$ is the 9-th cyclotomic polynomial and the roots of $f(X)=\Phi_9(X)$ are precisely the primitive 9-th roots of unity and thus $\deg f=\deg\Phi_9=\varphi(9)=3^2-3=6$. Thus, by Theorem 3.1, $f(X)=\Phi_9(X)=\mathrm{Irr}(\zeta_9,\mathbb{Q},X)$ where ζ_9 is a 9-th primitive root of unity. Hence $\mathbb{Q}(\zeta_9)$ is the splitting field of $f(X)$ and $[\mathbb{Q}(\zeta_9)\colon\mathbb{Q}]=\varphi(9)=6$.

g) This is similar to part (c) and thus $\mathbb{Q}(\sqrt[5]{7},\zeta_5)$ is the splitting field of X^5-7 over $\mathbb{Q}$ and by the tower law, noting that $\mathbb{Q}(\sqrt[5]{7})\cap\mathbb{Q}(\zeta_5)=\mathbb{Q}$, we have

$$[\mathbb{Q}(\sqrt{5}7,\zeta_5)\colon\mathbb{Q}]=[\mathbb{Q}(\sqrt[5]{7},\zeta_5)\colon\mathbb{Q}(\zeta_5)]\cdot[\mathbb{Q}(\zeta_5)\colon\mathbb{Q}]=5\cdot\varphi(5)=20.$$

□

Exercise 0.9. (Exercise 12)
Let K be a finite field with p^n elements. Show that every element of K has a unique p-th root in K.

Proof. Define $\varphi\colon K\to K$ by $\varphi(x)=x^p$, a Frobenius mapping. Hence, φ is an embedding. Furthermore, since K is finite, φ is surjective, and thus $\varphi\in\mathrm{Aut}(K)$. By surjectivity, for all $y\in K$, there exists some $x\in K$ such that $\varphi(x)=x^p=y$. Thus, $\sqrt[p]{y}=x\in K$. The uniqueness of the p-th root follows immediately from the injectivity of φ.

□

Exercise 0.10. (Exercise 13)
If the roots of a monic polynomial $f(X)\in k[X]$ in some splitting field are distinct, and form a field, then $\mathrm{char}\,k=p$ and $f(X)=X^{p^n}-X$ for some $n\geq1$.

Proof. Let $F=\{\alpha_1,\dots,\alpha_q\}$ be the set of distinct roots of $f(X)\in k[X]$, which is a field by assumption, where $|F|=q=\deg f$. Since F is a field with a finite number of elements, $\mathrm{char}\,F=p$ for some prime p. The splitting field of $f(X)$ over k is, by definition, the smallest subfield that contains k and all the roots of $f(X)$. But F is already a field that contains all the roots, so $k(F)\subseteq F$. Trivially, $F\subseteq k(F)$ and thus $k(F)=F\supset k$. Therefore, since $\mathrm{char}\,F=p$, we have $\mathrm{char}\,k=p$, proving the first part. Consider the Frobenius map $\varphi\colon F\to F$ given by $\varphi(x)=x^p$. By definition, φ is an embedding and, furthermore, since F is finite, a surjection. Thus, $\varphi\in\mathrm{Aut}(F)$. Then, there exists some $n\geq1$ such that $\varphi^n(x)=x^{p^n}=x$ for all $x\in F$. Hence, every element $x\in F$ satisfies the polynomial equation $X^{p^n}-X=0$. Thus,

$$F\subseteq\{x\in k^a\colon x^{p^n}-x=0\},$$

where k^a is an algebraic closure of k. The $\deg(X^{p^n}-X)=p^n$ and $\frac{d}{dx}\left[X^{p^n}-X\right]=-1\neq0$ which implies that $X^{p^n}-X$ is separable with exactly p^n distinct roots. By the above, F is a field containing

all the roots of $X^{p^n} - X$ and $|F| = q$, so we must have $q = p^n$. Hence, $F = \{x \in k^a : x^{p^n} - x = 0\}$. By definition, F was the set of roots of the monic polynomial $f(X) \in k[X]$ and it is also exactly the set of roots of $X^{p^n} - X$.
Therefore, $f(X)$ and $X^{p^n} - X$ are both monic and separable with the same set of roots, F. Hence, it follows that they must be equal. Thus, $f(X) = X^{p^n} - X \in k[X]$.

□

Exercise 0.11. (Exercise 14)
Let $\operatorname{char} K = p$. Let L be a finite extension of K, and suppose $[L\colon K]$ prime to p. Show that L is separable over K.

Proof. Let $L = K(\alpha_1, \ldots, \alpha_n)$ be a finite extension of K and thus L is also algebraic. Assume $([L\colon K], p) = 1$. Let $\alpha \in L$. By way of contradiction, suppose α is inseparable. Then the derivative of $\operatorname{Irr}(\alpha, K, X)$ is 0 and since $\operatorname{char} K = p$, this implies that the degree of the irreducible polynomial of α over K is divisible by p. Since $[K(\alpha)\colon K] = \deg \operatorname{Irr}(\alpha, K, X)$, we have $p \Big| [K(\alpha)\colon K]$. Furthermore, we have the following tower of fields,

$$K \subseteq K(\alpha) \subseteq L$$

Thus, $[L\colon K] = [L\colon K(\alpha)] \cdot [K(\alpha)\colon K]$. Therefore, $[K(\alpha)\colon K] \Big| [L\colon K]$. Then from $p \Big| [K(\alpha)\colon K]$, we obtain $[K(\alpha)\colon K] = pr$ with $r \in \mathbb{Z}$. Hence,

$$pr \Big| [L\colon K] \Rightarrow [L\colon K] = (pr)s = p(rs) \Rightarrow p \Big| [L\colon K] \quad (s \in \mathbb{Z})$$

This is a contradiction! We assumed $([L\colon K], p) = 1$. Thus, α is separable and since α was arbitrary, every $\alpha \in L$ is separable. Therefore, $L \supset K$ is separable.

□

Exercise 0.12. (Exercise 15)
Suppose $\operatorname{char} K = p$. Let $a \in K$. If a has no p-th root in K, show that $X^{p^n} - a$ is irreducible in $K[X]$ for all positive integers n.

Proof. Assume that $\operatorname{char} K = p$ and $a \in K$. Suppose that a has no p-th root in K, i.e., there is no $b \in K$ such that $b^p = a$. Let $n \geq 1$. By way of contradiction, suppose $f(X) = X^{p^n} - a$ is reducible in $K[X]$. Let α be a root of $f(X)$ in an algebraic closure K^a. Then $\alpha^{p^n} = a \in K$. Note that $f'(X) = 0$, so $f(X)$ is inseparable. In characteristic p, we have

$$X^{p^n} - a = (X - \alpha)^{p^n}$$

in $K^a[X]$. Thus $f(X)$ has a unique root α (of multiplicity p^n) in K^a.
Since $f(X)$ is reducible over K, there exist $g(X), h(X) \in K[X]$ with $\deg g, \deg h \geq 1$ such that $f(X) = g(X)h(X)$. Without loss of generality, suppose α is a root of $g(X)$ and $\deg g(X) = p^m$ with $m < n$ (since the degree of the minimal polynomial of α over K must divide p^n and be strictly less than p^n).
Define $\beta = \alpha^{p^m}$. Then

$$\beta^{p^{n-m}} = \left(\alpha^{p^m}\right)^{p^{n-m}} = \alpha^{p^n} = a.$$

So a is a p^{n-m}-th power in the extension $K(\alpha)/K$. But since $[K(\alpha) : K] = p^m$ and $n - m \geq 1$, and the Frobenius is purely inseparable, the assumption that $X^{p^n} - a$ is reducible forces a to become a lower p-power in a smaller purely inseparable extension.
Repeating this process (or using induction), we eventually obtain that a is a p-th power in K, i.e., there exists $b \in K$ with $b^p = a$. This contradicts the hypothesis that a has no p-th root in K.
Therefore $X^{p^n} - a$ is irreducible in $K[X]$ for every positive integer n. □

Exercise 0.13. (Exercise 16)
Let $\operatorname{char} K = p$. Let α be algebraic over K. Show that α is separable if and only if $K(\alpha) = K(\alpha^{p^n})$ for all positive integers n.

Proof. Let $\operatorname{char} K = p$ and α be algebraic over K with $\operatorname{Irr}(\alpha, K, X) = f(X)$ and $\deg f(X) = d = [K(\alpha) \colon K]$.
($\implies$) Assume α is separable over K. Then $f'(X) \neq 0$ and thus $p \nmid d$. Fix any $n \in \mathbb{Z}^+$ and set $\beta = \alpha^{p^n}$. Hence, $\beta \in K(\alpha)$ and thus $K(\beta) \subseteq K(\alpha)$. Observe that α is a root of $X^{p^n} - \beta \in K(\beta)[X]$. Thus, $g(X) = \operatorname{Irr}(\alpha, K(\beta), X)$ divides $X^{p^n} - \beta$ and since $\operatorname{char} K = p$, $\deg g(X) = p^k \leq p^n$. Hence, $[K(\alpha) \colon K(\beta)] = p^k \leq p^n$. From the tower,

$$K \subseteq K(\beta) \subseteq K(\alpha),$$

we have

$$[K(\alpha) \colon K] = [K(\alpha) \colon K(\beta)] \cdot [K(\beta) \colon K]$$

Thus, $[K(\alpha) \colon K(\beta)] \Big| [K(\alpha) \colon K]$. Hence, we have $p^k \big| d$, $p^k \leq p^n$, and $p \nmid d$. Thus, we must have $p^k = 1$. Therefore, $[K(\alpha) \colon K(\beta)] = 1 \Rightarrow K(\alpha) = K(\beta) = K(\alpha^{p^n})$.
($\impliedby$) Assume $K(\alpha) = K(\alpha^{p^n})$ with $n \in \mathbb{Z}^+$. By way of contradiction, suppose α is inseparable. Then $f'(X) = 0$ and thus $f(X) = g(X^p)$ for some $g(X) \in K[X]$ with $\deg g(X) = \frac{d}{p} \in \mathbb{Z}$. Since $[K(\alpha) \colon K] = d \geq 2$ ($d = 1$ is trivial), we have $\frac{d}{p} < d$. By definition, we have $f(\alpha) = 0$ and thus $g(\alpha^p) = 0$ and since $g(X)$ is irreducible, we must have $g(X) = \operatorname{Irr}(\alpha^p, K, X)$. Thus, $[K(\alpha^p) \colon K] = \deg g(X) = \frac{d}{p}$. By assumption, taking $n = 1$, $K(\alpha) = K(\alpha^p)$

$$d = [K(\alpha) \colon K] = [K(\alpha^p) \colon K] = \frac{d}{p}$$

Either from our previous remark of $\frac{d}{p} < d$ or from $d(p-1) = 0$, we see that there is clearly a contradiction. Thus, α is separable over K.

□

Exercise 0.14. (Exercise 17)
Prove that the following tow properties are equivalent:

a) Every algebraic extension of K is separable.

b) Either $\operatorname{char} K = 0$, or $\operatorname{char} K = p$ and every element of K has a p-th root in K.

Proof. ($a \implies b$) Assume that every algebraic extension of K is separable. There are two possible characteristics for K, either 0 or some prime p. If $\operatorname{char} K = 0$, then all irreducibles in $K[X]$ are separable. Thus, in this case, the implication is trivial. Now suppose $\operatorname{char} K = p$. By way of contradiction, suppose that there exists some $a \in K$ such that $\sqrt[p]{a} \notin K$. Then the polynomial $X^p - a \in K[X]$ is irreducible. Observe that $\frac{d}{dX}[X^p - a] = 0$, thus all the roots of $X^p - a$ are multiple. Hence $K(\alpha)$, with α a root of $X^p - a$, would be an algebraic extension that is not separable (α is a multiple root), contradicting our assumption. Thus, every element of K has a p-th root in K.
($b \impliedby a$) Assume $\operatorname{char} K = 0$. In characteristic 0, all irreducible polynomials over K are separable and thus every algebraic extension is separable. Now suppose $\operatorname{char} K = p$ and every element of K has a p-th root in K. Let α be any algebraic element over K. By *Exercise 16*, α is separable if and only if $K(\alpha) = K(\alpha^{p^n})$ for all $n \in \mathbb{Z}^+$. Since every element of K has a p-th root by assumption, $K(\alpha) = K(\alpha^{p^n})$ for all n. Hence, α is separable. By induction, every algebraic extension of K is separable.

□

Exercise 0.15. (Exercise 18)
Show that every element of a finite field can be written as a sum of two squares in that field.

Proof. Let k be a finite field with p^n elements for some prime p and $n \geq 1$. Then $k^\times$ is cyclic with the generator $x \in k$ such that $k^\times = \langle x \rangle$ and $|k^\times| = p^n - 1$. The perfect squares in $k^\times$ are precisely the elements $y \in k^\times$ such that $y = x^2$ for some $x \in k^\times$. We denote the perfect squares by $(k^\times)^2 \subseteq k^\times$, and thus x^2 generates the perfect squares, i.e. $(k^\times)^2 = \langle x^2 \rangle$. Observe that the squares of k, denoted k^2, require us to "add" 0 to $(k^\times)^2$, (since $0^2 = 0$) thus $|k^2| = \frac{p^n-1}{2} + 1 = \frac{p^n+1}{2}$. Let $y \in k$. Consider

$$S := \left\{ y - (x^2)^n \colon n = 0, 1, \ldots, \frac{p^n - 1}{2} \right\} \cup \{0\}.$$

Note that $(x^2)^n \in (k^\times)^2 = \langle x^2 \rangle$ and we have $|S| = |k^2| = \frac{p^n+1}{2}$ with $S, k^2 \subseteq k$. Then observe that

$$\begin{aligned}
|S| + |k^2| &= \frac{p^n+1}{2} + \frac{p^n+1}{2} \\
&= \frac{2p^n+2}{2} \\
&= p^n + 1 \\
&= |k| + 1 > |k|
\end{aligned}$$

Therefore by the Pigeonhole principle, S contains a perfect square ($S \cap k^2 \neq \varnothing$) and thus there exists some $z = a^2 \in k^2$ such that for some $w = x^n \in k^\times$, we have

$$\begin{aligned}
y - w^2 =& a^2 \\
y = a^2 + w^2
\end{aligned}$$

as desired.

□

Exercise 0.16. (Exercise 19)
Let E be an algebraic extension of F. Show that every subring of E which contains F is actually a field. Is this necessarily true if E is not algebraic over F? Prove or give a counterexample.

Proof. Let $K \subseteq E$ be a subring such that $F \subseteq K$. Note that if $K = E$ or $K = F$, then we are done. Thus, suppose that $K \subsetneq E$ and $F \subsetneq K$. Hence, K is a vector space over F. Let $a \in K$. Consider the map

$$\varphi_a \colon K \to K$$

given by $x \mapsto ax$. Thus, φ_a is linear and since K is contained in a field E, K is entire. Hence, if $x \in \ker \varphi_a$, then $ax = 0 \Rightarrow x = 0$. Then φ_a is injective and is a map between itself (i.e. between vector spaces of equal dimension) and thus φ_a is surjective. Since $1 \in K$, there exists some $y \in K$ such that $\varphi_a(y) = ay = 1$, that is, a is a unit and $a \in K$ was arbitrary, so K is a field.
We show that this is not necessarily true if E is not algebraic over F via a counterexample.
Let x be transcendental over $\mathbb{Q}$. Hence, $\mathbb{Q}[x] \cong \mathbb{Q}[X]$ via the evaluation map. Since $\mathbb{Q} \subseteq \mathbb{Q}[X]$, by the above, we have $\mathbb{Q} \subseteq \mathbb{Q}[x]$. We also know that for any $y \in \mathbb{Q}(x)$, we can write $y = f(x)/g(x)$ with $g(x) \neq 0$. Hence, $\mathbb{Q}[x] \subseteq \mathbb{Q}(x)$. Thus, by definition, $\mathbb{Q}$ and $\mathbb{Q}(x)$ are fields. Observe that $x \in \mathbb{Q}[x]$, but $x^{-1} = 1/x \notin \mathbb{Q}[x]$ ($1/x \in \mathbb{Q}(x)$). Thus, $\mathbb{Q}[x]$ is not closed under inverses and is therefore not a field.

□

Exercise 0.17. (Exercise 20)

a) Let $E = F(x)$ where x is transcendental over F. Let $K \neq F$ be a subfield of E which contains F. Show that x is algebraic over K.

b) Let $E = F(x)$. Let $y = f(x)/g(x)$ be a rational function, with relatively prime polynomials $f, g \in F[x]$. Let $n = \max(\deg f, \deg g)$. Suppose $n \geq 1$. Prove that

$$[F(x) \colon F(y)] = n.$$

Proof. a) By definition of x transcendental over F, we have $F[X] \cong F[x]$ given by given by $f(X) \mapsto f(x)$. By assumption $F \subsetneq K \subseteq E = F(x)$. Let $y \in K - F$. Then since $K \subseteq F(x)$, $y \in F(x)$ so there exists some $f(x), g(x) \in F[x]$ with $g(x) \neq 0$ such that $y = f(x)/g(x)$. Hence

$$f(x) - yg(x) = 0.$$

By lifting the evaluation map, set $m(X) = f(X) - yg(X) \in F(y)[X]$. Hence $m(x) = 0$, and thus $F(x)$ is an algebraic extension of $F(y)$. Since $F(y) \subseteq K \subseteq F(x)$, it follows that $F(x)$ is algebraic over K and thus x is algebraic over K.

b) Consider $m(X) = f(X) - yg(X) \in F(y)[X]$ from part (a). We wish to show that $m(X) = \operatorname{Irr}(x, F(y), X)$. Thus, we proceed by showing that $m(X)$ is irreducible. By Gauss Lemma, irreducibility of $m(X)$ over $F(y)[X]$ is the same as irreducibility over $F[y][X] = F[y, X]$. We see that $m(X)$ is linear as a polynomial in y. Thus, it is irreducible over $F[y, X]$. Therefore $[F(x) \colon F(y)] = \deg \operatorname{Irr}(x, F(y), X) = \deg m(X)$. Observe that the factor of y on $g(X)$ ensures that no higher power can cancel with any terms of $f(X)$. Thus, counting degrees tells us that $\deg m(X) = \max(\deg f, \deg g) = n$. Hence, $[F(x) \colon F(y)] = n$.

□

Exercise 0.18. (Exercise 22)
Let k be a finite field with q elements. Let $f(X) \in k[X]$ be irreducible. Show that $f(X)$ divides $X^{q^n} - X$ if and only if $\deg f$ divides n. Show the multiplication formula

$$X^{q^n} - X = \prod_{d|n} \prod_{f_d \text{ irr}} f_d(X),$$

where the inner product is over all irreducible polynomials of degree d with leading coefficient 1. Counting degrees, show that

$$q^n = \sum_{d|n} d\psi(d),$$

where $\psi(d)$ is the number of irreducible polynomials of degree d. Invert by Exercise 21 and find that

$$n\psi(n) = \sum_{d|n} \mu(d) q^{n/d}.$$

Proof. **PART OF PROOF IN NOTES!!!**
???

□

Exercise 0.19. (Exercise 26)
Let k be a field, $f(X)$ an irreducible polynomial in $k[X]$, and let K be a finite normal extension of k. If g, h are monic irreducible factors of $f(X)$ in $K[X]$, show that there exists an automorphism σ of K over k such that $g = h^\sigma$. Give an example when this conclusion is not valid if K is not normal over k.

Proof. Let $f(X) = g(X)h(X) \in K[X]$ with $g(X), h(X)$ monic irreducible polynomials over K. Let k^a be an algebraic closure. Then we can write

$$g(X) = \prod (X - \alpha_i) \quad \text{and} \quad h(X) = \prod (X - \beta_i)$$

in $k^a[X]$. Define $\sigma \colon k^a \to k^a$ by $\alpha_1 \mapsto \beta_1$. Then since K is normal, $\sigma|_K$ induces an automorphism of K given by $\sigma|_K = \sigma(\alpha_1) = \beta_1$. Thus, by our assumption that $g(X), h(X) \in K[X]$ are irreducible, $h(X)$ is the minimal polynomial of $\sigma(\alpha_1)$. Inductively, it follows that $h(X)$ is the minimal polynomial of $\sigma(\alpha_i)$ for all roots α_i. Hence $h = g^\sigma$ (the problem statement says to prove that $h^\sigma = g$, but this proof is equivalent, we just defined σ differently).
By the above argument and Theorem 3.1, there is a bijection between the roots of $g(X)$ and the roots of $h(X)$, so $\deg g(X) = \deg h(X)$.
For a counter-example, where there is no normality, consider $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$. Then, we can write

$$f(X) = X^3 - 2 = (X - \sqrt[3]{2})(X^2 + \sqrt[3]{2} + \sqrt[3]{4}) \in \mathbb{Q}(\sqrt[3]{2})[X].$$

Both factors of $f(X)$ are monic and irreducible; however, their degrees differ. Thus, there is no way to form an automorphism $\sigma \colon K \to K$ in an algebraic closure since there is no bijection between their roots.

□